

Xinyu Shan

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Education

Lanzhou University	Sep 2017 - Jun 2020
Computational Mathematics Master School of Mathematics and Statistics	Lanzhou
Fudan University	Sep 2021 - Jun 2026
Statistics Doctor School of Data Science	Shanghai

Research Interests

Eigenvalue Problems

Researching effective algorithms for solving eigenvalue problems, with a focus on contour integral methods, LOBPCG algorithms, and subspace iterative methods

Randomized Algorithms

Exploring the applications of randomized algorithms in matrix equations and matrix decompositions, including randomized projection, randomized coordinate descent, and Kaczmarz algorithms

Publications

A contour integral-based algorithm for computing generalized singular values	
Yuqi Liu , Xinyu Shan , Meiyue Shao	Journal of Computational and Applied Mathematics
A structure-preserving LOBPCG algorithm for the Bethe-Salpeter eigenvalue problem	
Xinyu Shan , Meiyue Shao	Under review
A structure-preserving LOBPCG method for computing several of the largest generalized singular triplets	
Xinyu Shan	Under review

Software Copyrights

Lot Acceptance Calculator V 1.0	
Xinyu Shan, Meiyue Shao, Zhenyu Xu, Shuqin Zhang	Registration No.: 2024SR1129341

Presentations

Structure-Preserving LOBPCG Algorithm for Solving the Bethe-Salpeter Eigenvalue Problem	May 2026
Applied Math Ph.D. Seminar	Shanghai
Computing Generalized Singular Value Decomposition Through Contour Integratio	May 2024
SIAM LA 2024 Minisymposium	Paris
Computing Generalized Singular Value Decomposition Through Contour Integration	Oct 2023
CSIAM 2023 Minisymposium	Kunming

Project Experience

Criteria for Lot Acceptance of Composite Materials	Aug 2022 - Aug 2024
- Develop a model for establishing the criteria for lot acceptance of composite materials - Develop corresponding calculation programs and complete the migration of core algorithms into C code - Draft project documentations, including software copyright applications and final research reports	

Honors & Awards

Travel Awards | SIAM Conference on Applied Linear Algebra | May 2025
Meritorious Winner | Interdisciplinary Contest In Modeling | Jan. 2017